

ENME608 Mechanisms

Position Analysis (Norton Ch. 4)

Aaron Taylor — 24232594

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Position Analysis — Overview

Position analysis answers the question: *given the link lengths and the input position, where is every link and point in the mechanism?* It is the foundation for velocity and acceleration analysis — the same vector loop is differentiated once for velocity and twice for acceleration.

Two equivalent approaches:

- **Graphical** — draw the linkage to scale, measure angles. Fast for a check, limited accuracy (*the graphic technique is replaced by the analytical method in this course*).
- **Analytical (vector loop)** — write a closed vector polygon, resolve into real/imaginary (or x/y) components, solve the two scalar equations. Exact, and the only method examinable here.

Exam focus. Always start by classifying the mechanism (four-bar? slider-crank? inverted?) and labelling which variables are **known** (inputs, constant link lengths) and which are **unknown** (outputs). The correct standard formula then follows directly from the lecture slides.

Open-book: Lec 3–4, Tutorial 4.

The Vector Loop Equation

Represent each link by a position vector $\mathbf{R} = R e^{j\theta}$ in the complex plane, where R is the (constant) link length and θ its angle from the global $+X$ axis. Walk around the closed kinematic chain; the vectors must sum to zero:

$$\mathbf{R}_2 + \mathbf{R}_3 - \mathbf{R}_4 - \mathbf{R}_1 = 0.$$

Using Euler's identity $e^{j\theta} = \cos \theta + j \sin \theta$ and separating real and imaginary parts gives **two scalar equations**:

$$\underbrace{a \cos \theta_2 + b \cos \theta_3 - c \cos \theta_4 - d \cos \theta_1}_{\text{real } (x)} = 0, \quad \underbrace{a \sin \theta_2 + b \sin \theta_3 - c \sin \theta_4 - d \sin \theta_1}_{\text{imag } (y)} = 0.$$

Here $a = L_2$ (crank), $b = L_3$ (coupler), $c = L_4$ (rocker/output), $d = L_1$ (ground, usually $\theta_1 = 0$).

Exam focus. The **direction you draw each arrow** fixes the signs in the loop equation. Once the arrows are set, the equations are unique. If you reverse an arrow, that term changes sign.

Four-Bar Linkage — Closed-Form Solution

With $\theta_1 = 0$ and the loop above, Norton's standard solution introduces the constants:

$$K_1 = \frac{d}{a}, \quad K_2 = \frac{d}{c}, \quad K_3 = \frac{a^2 - b^2 + c^2 + d^2}{2ac},$$
$$K_4 = \frac{d}{b}, \quad K_5 = \frac{c^2 - d^2 - a^2 - b^2}{2ab}.$$

Output angle θ_4 (solve the quadratic in $\tan(\theta_4/2)$):

$$A \tan^2 \frac{\theta_4}{2} + B \tan \frac{\theta_4}{2} + C = 0, \quad \theta_4 = 2 \arctan \left(\frac{-B \mp \sqrt{B^2 - 4AC}}{2A} \right),$$

$$A = \cos \theta_2 - K_1 - K_2 \cos \theta_2 + K_3, \quad B = -2 \sin \theta_2, \quad C = K_1 - (K_2 + 1) \cos \theta_2 + K_3.$$

Coupler angle θ_3 :

$$D \tan^2 \frac{\theta_3}{2} + E \tan \frac{\theta_3}{2} + F = 0, \quad \theta_3 = 2 \arctan \left(\frac{-E \mp \sqrt{E^2 - 4DF}}{2D} \right),$$

$$D = \cos \theta_2 - K_1 + K_4 \cos \theta_2 + K_5, \quad E = -2 \sin \theta_2, \quad F = K_1 + (K_4 - 1) \cos \theta_2 + K_5.$$

Exam focus. The $\mp$ gives the **two assembly modes**: the minus sign is the **open** configuration, the plus sign the **crossed**. The discriminant must be ≥ 0 for the linkage to assemble at that θ_2 — if it is negative, the mechanism cannot reach that input angle (a limit position).

Slider-Crank (Crank-Slider) — Closed-Form Solution

For the offset slider-crank (crank $a = L_2$, coupler $b = L_3$, offset c , slider on a line parallel to X):

$$\theta_3 = \arcsin \left(\frac{a \sin \theta_2 - c}{b} \right), \quad d = a \cos \theta_2 - b \cos \theta_3,$$

where d is the slider position. The two roots of the arcsin (θ_3 and $180^\circ - \theta_3$) again correspond to the open and crossed branches.

Inverted slider-crank (slider block pivots on a rotating link): the block angle θ_3 is tied to the rotating link angle θ_4 by a constant offset γ (usually 90°): $\theta_3 = \theta_4 + \gamma$. This relation reduces the unknowns so the loop can be solved

Open-book: Lec 4, inverted slider-crank slide.

Transmission Angle

The **transmission angle** μ is the angle between the coupler (link 3) and the output link (link 4) at their common joint. It measures how effectively force is transmitted to the output:

$$\mu = |\theta_4 - \theta_3| \quad (\text{take the acute value } \leq 90^\circ).$$

- $\mu = 90^\circ$: **ideal** — all coupler force drives the output tangentially.
- $\mu \rightarrow 0^\circ$ (or 180°): force is directed along the output link; mechanism is near a **toggle**/lock-up. Design rule of thumb: keep $\mu \geq 40^\circ$ throughout the cycle.

Law-of-cosines method (no full position solution needed). Build the diagonal S from the crank pivot O_2 to the output joint B using the two triangles that share it.

1. Triangle O_2-A-O_4 (sides: crank a , ground d , included angle θ_2 measured *from the ground line*):

$$S^2 = a^2 + d^2 - 2ad \cos \theta_2.$$

2. Triangle $A-B-O_4$ (sides: coupler b , output c , included angle μ):

$$S^2 = b^2 + c^2 - 2bc \cos \mu \Rightarrow \cos \mu = \frac{b^2 + c^2 - S^2}{2bc}.$$

If $\mu > 90^\circ$, the transmission angle quoted is $180^\circ - \mu$.

Exam focus. Extreme transmission angles occur when the crank is **collinear with the ground link** ($\theta_2 = 0^\circ$ and $\theta_2 = 180^\circ$), giving $S = |d - a|$ and $S = d + a$. Evaluate $\cos \mu$ at both to bracket $\mu_{\min}$ and $\mu_{\max}$.

Motion-Limiting (Toggle) Positions

A **toggle** (limit) position occurs when two of the moving links become **collinear**, so the mechanism momentarily cannot be driven further (it would need infinite force, or it can flip either way).

- **Crank-rocker / rocker output limit:** coupler (3) and rocker (4) become collinear $\Rightarrow$ effective length A to O_4 is $b + c$ (extended) or $|b - c|$ (folded). Solve for the crank angle θ_2 from triangle O_2-A-O_4 :

$$\cos(\angle AO_2O_4) = \frac{a^2 + d^2 - (b \pm c)^2}{2ad}.$$

- **Driver (crank) toggle:** crank (2) and coupler (3) become collinear $\Rightarrow$ distance O_2 to B is $a + b$ or $|a - b|$.

Two types of motion-limiting condition (from lecture):

1. **Toggle position** — collinearity of crank and coupler; can occur for a crank-rocker. Inertia is needed to carry the mechanism through.
2. **Stationary configuration** — collinearity between coupler and output, the extreme positions of the rocker's swing.

Quick-Return and Timing Ratio

A linkage whose output moves faster on the return stroke than the forward stroke is a **quick-return** mechanism. Defining the crank rotation for the forward stroke α and return stroke β (with $\alpha + \beta = 360^\circ$):

$$\text{Time ratio TR} = \frac{\alpha}{\beta}, \quad \text{Construction angle } \delta = 180^\circ - \alpha = \beta - 180^\circ.$$

Knowing any one of $\{\alpha, \beta, \text{TR}, \delta\}$ fixes the others, from which the linkage geometry (e.g. crank length) is designed

Open-book: Lec 4, quick-return slide.

Tutorial Solutions (Tutorial 4 — Position B)

P3-15 — Transmission Angle and Toggle of a Non-Grashof Four-Bar

3-15 Figure P3-4 shows a non-Grashof fourbar linkage that is driven from link O_2A . All dimensions are in centimeters (cm).

- Find the transmission angle at the position shown.
- Find the toggle positions in terms of angle AO_2O_4 .
- Find the maximum and minimum transmission angles over its range of motion by graphical techniques.

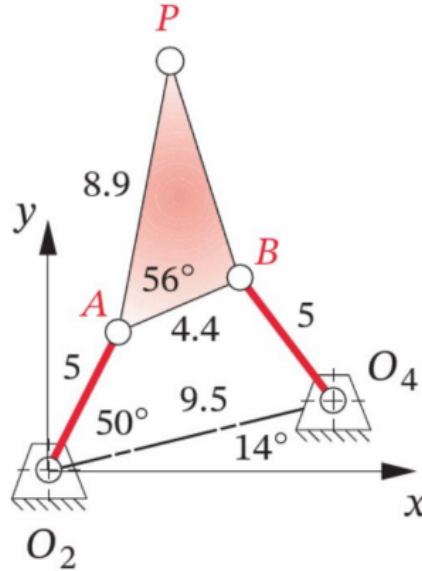


Figure P3-4: non-Grashof four-bar driven from link O_2A (dimensions in cm).

Given:

- Crank $a = L_2 = O_2A = 5$; coupler $b = L_3 = AB = 4.4$; output $c = L_4 = O_4B = 5$; ground $d = L_1 = O_2O_4 = 9.5$ (cm).
- Position shown: crank at $\theta_2 = 50^\circ$ from the ground line.

Find:

(a) transmission angle at the position shown; (b) the toggle position(s) in terms of $\angle AO_2O_4$; (c) the maximum and minimum transmission angles over the range of motion.

Solution:

(a) Diagonal S from triangle O_2A-O_4 :

$$S^2 = a^2 + d^2 - 2ad \cos \theta_2 = 5^2 + 9.5^2 - 2(5)(9.5) \cos 50^\circ = 115.25 - 61.06 = 54.19 \Rightarrow S = 7.36.$$

Then in triangle $A-B-O_4$:

$$\cos \mu = \frac{b^2 + c^2 - S^2}{2bc} = \frac{4.4^2 + 5^2 - 54.19}{2(4.4)(5)} = \frac{-9.83}{44} = -0.223 \Rightarrow \mu = 102.9^\circ.$$

Since $\mu > 90^\circ$, the transmission angle is $180^\circ - 102.9^\circ = \boxed{77.1^\circ}$.

(b) The driving (crank) toggle occurs when coupler and output are collinear, $AO_4 = b + c = 9.4$, in triangle O_2A-O_4 :

$$\cos(\angle AO_2O_4) = \frac{a^2 + d^2 - (b + c)^2}{2ad} = \frac{25 + 90.25 - 88.36}{2(5)(9.5)} = 0.283 \Rightarrow \angle AO_2O_4 = \boxed{73.55^\circ}.$$

(The folded case $|b-c| = 0.6$ gives $\cos > 1$: no second toggle, consistent with the non-Grashof geometry.)

(c) At the toggle, coupler and output are collinear so $\mu = 0^\circ$; the linkage also passes through $\mu = 90^\circ$. Hence the transmission angle ranges $\boxed{0^\circ \text{ to } 90^\circ}$.

P3-22 — Max/Min Transmission Angle of a Driving-Stage Four-Bar

*3-22 Find the maximum and minimum transmission angles of the fourbar driving stage (links L_1, L_2, L_3, L_4) in Figure P3-7 to graphical accuracy.

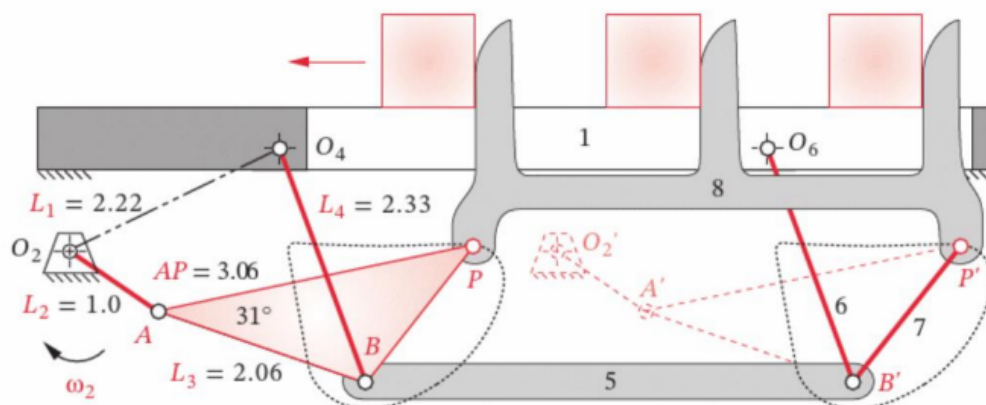


FIGURE P3-7

Figure P3-7: four-bar driving stage, $L_1 = 2.22$, $L_2 = 1.0$, $L_3 = 2.06$, $L_4 = 2.33$.

Given:

$$a = L_2 = 1.0, b = L_3 = 2.06, c = L_4 = 2.33, d = L_1 = 2.22.$$

Find:

Maximum and minimum transmission angles.

Assuming:

Grashof crank-rocker (shortest link = crank = 1.0; $S + L = 1.0 + 2.33 = 3.33 < P + Q = 2.06 + 2.22 = 4.28$), so the crank fully rotates and the extremes occur at $\theta_2 = 0^\circ, 180^\circ$.

Solution:

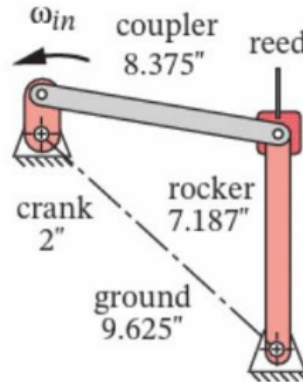
$$\text{At } \theta_2 = 0^\circ: S = d - a = 1.22,$$

$$\cos \mu = \frac{b^2 + c^2 - S^2}{2bc} = \frac{2.06^2 + 2.33^2 - 1.22^2}{2(2.06)(2.33)} = 0.8525 \Rightarrow \mu_{\min} = \boxed{31.51^\circ}.$$

At $\theta_2 = 180^\circ$: $S = d + a = 3.22$, giving $\cos \mu = -0.073$, $\mu = 94.2^\circ$, i.e. 85.8° . As θ_2 sweeps between these, μ passes through $\boxed{90.0^\circ}$ (the maximum effective transmission angle).

P3-23 — Grashof Condition and Transmission Angles of a Loom Drive

- *3-23 Figure P3-8 shows a fourbar linkage used in a power loom to drive a comblike reed against the thread, “beating it up” into the cloth. Determine its Grashof condition and its minimum and maximum transmission angles to graphical accuracy.



[View as a video](http://www.designof-machinery.com/DOM/loom_laybar_drive.avi)

http://www.designof-machinery.com/DOM/loom_laybar_drive.avi

FIGURE P3-8

Figure P3-8: power-loom four-bar. crank 2", coupler 8.375", rocker 7.187", ground 9.625".

Given:

crank $a = 2$, coupler $b = 8.375$, rocker $c = 7.187$, ground $d = 9.625$ (in).

Find:

Grashof condition; minimum and maximum transmission angles.

Solution:

Grashof: shortest $S = 2$, longest $L = 9.625$; $S + L = 11.625 < P + Q = 8.375 + 7.187 = 15.562$. Condition satisfied, and since the shortest link is the crank, this is a **Grashof crank-rocker**.

At $\theta_2 = 0^\circ$: $S = d - a = 7.625$,

$$\cos \mu = \frac{8.375^2 + 7.187^2 - 7.625^2}{2(8.375)(7.187)} = 0.529 \Rightarrow \mu_{\min} = \boxed{58.08^\circ}.$$

At $\theta_2 = 180^\circ$: $S = d + a = 11.625$, $\cos \mu = -0.111$, $\mu = 96.4^\circ \rightarrow 83.6^\circ$; the sweep again reaches $\mu_{\max} = \boxed{90.0^\circ}$.

Worked Example — Four-Bar Vector-Loop Solution (Row a, feeds Tut 5 & 6)

This is the open-configuration position solution for the linkage used throughout the velocity (Tut 5, P6-5) and acceleration (Tut 6, P7-4) tutorials. It is worth committing the result to memory as a self-check for those problems.

Given:

$L_1 = d = 6$, $L_2 = a = 2$, $L_3 = b = 7$, $L_4 = c = 9$, $\theta_2 = 30^\circ$ (Table P4-1, Row a).

Find:

θ_3, θ_4 (open configuration).

Solution:

Constants:

$$K_1 = \frac{6}{2} = 3, \quad K_2 = \frac{6}{9} = 0.667, \quad K_3 = \frac{4-49+81+36}{2(2)(9)} = 2.0,$$

$$K_4 = \frac{6}{7} = 0.857, \quad K_5 = \frac{81-36-4-49}{2(2)(7)} = -0.286.$$

For θ_4 : $A = -0.711$, $B = -1.0$, $C = 3.557$, discriminant = 11.12,

$$\theta_4 = 2 \arctan\left(\frac{1.0 - 3.335}{2(-0.711)}\right) = \boxed{117.3^\circ}.$$

For θ_3 : $D = -1.677$, $E = -1.0$, $F = 2.591$, discriminant = 18.38,

$$\theta_3 = 2 \arctan\left(\frac{1.0 - 4.287}{2(-1.677)}\right) = \boxed{88.8^\circ}.$$

Check (loop closure, y): $a \sin \theta_2 + b \sin \theta_3 = 1.0 + 6.999 = 8.00 \approx c \sin \theta_4 = 9 \sin 117.3^\circ = 8.00 \checkmark$